MIL-P-11678B(MU)
2 October 1968
SUPERSEDING
MIL-S-11678A(CmlC)
1 April 1955

MILITARY SPECIFICATION

PROJECTILE, 90-MM, SMOKE, WP, M313;

FILLING AND ASSEMBLY

1. SCOPE

1.1 Scope. This specification covers the white phosphorus (WP) filling and assembly of the M313 projectile without fuzing components.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY MUNITIONS COMMAND

ORDNANCE CORPS

75-14-501 - Shell for Cartridge, 90MM, Smoke, WP, M313: Loading Assembly and Marking Diagram.

PICATINNY ARSENAL

8837835 - Pallet for Projectile Metal Parts.
8861520 - Marking Diagram for Pallet for Projectiles 75MM-120MM.

FSC 1315

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 75-14-501, or on the subsidiary drawings, shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile shall conform to the specifications and drawings listed on Drawing 75-14-501 and subsidiary drawings.

3.2 Filling and assembly.

3.2.1 Filling. The projectile, clean and free of foreign matter, shall be filled with molten WP conforming to MIL-C-215 (see 6.2 and 6.3) at a temperature of 140° ± 10°F. When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After test for leakage, the WP shall be solidified with the projectile in a filling-end-up position.

3.2.2 Filling height. There shall be approximately a 7 to 10 percent void, by total volume, existing above the fill in the filled and closed projectile. The depth of water remaining above the WP in the filled and closed projectile shall not exceed 1/8 inch, when applicable. The projectile shall be radiographed in a vertical, filling-end-up position while the temperature of the WP is 130° ± 10°F when tested as specified in 4.2.4.1, to determine proper fill height, and location of burster casing.

3.2.3 Assembly. The projectile shall be assembled in accordance with Drawing 75-14-501.

3.3 Cleaning after closing. The exterior surface of the filled and closed projectile and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, WP, or other foreign matter (see 6.4).

3.4 Leakage. The closed and cleaned projectile shall not leak when maintained at a temperature of 160° ± 5°F, for a period of 6 hours. Alternate method: the projectile shall not leak when maintained at a temperature of 207° to 212°F for a minimum period of 15 minutes. The projectile shall be tested as specified in 4.2.4.2.

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3.5 Workmanship. The projectile shall be free of all foreign matter (oil, grease, contamination, etc.) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of projectiles, of one interfix number, filled by the same filling installation, under essentially the same conditions. However, no more than one manufacturer or interfix number of shells and burster wells shall be represented in any one lot of projectiles.

4.2.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample projectiles shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each filled and closed projectile shall be inspected for the critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects.

(a) Projectile, 90MM, Smoke, WP, M313; loading assembly and marking diagram (Drawing 75-14-501).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.2
2	WP contamination (3.3)	
Major:	AQL 1.0 percent defective	
*101	Filling excessive	4.2.4.1
*102	Excess water	4.2.4.1
*103	Filling insufficient	4.2.4.1
104	Component missing or incorrectly assembled	
105	Burster casing location incorrect	4.2.4.1
106	Bourrelet diameter incorrect after protective finish and marking application	
107	Marking incorrect	
108	Protective finish missing	
109	Workmanship (3.5)	
Minor:	AQL 2.5 percent defective	
201	Marking missing or illegible	
202	Adapter thread protector missing	

*Note: Sampling for inspection for these characteristics, after first acceptable lot, shall be in accordance with MIL-STD-105, general inspection level II.

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4.2.3.3 Classification of defects (continued).

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Pallets incorrect
102	Protective separators missing
103	Strapping loose or missing
104	Marking incorrect
Minor:	AQL 10.0 percent defective
201	Marking illegible

4.2.4 Tests.

4.2.4.1 Filling height. The filled and closed projectile shall be radiographed while the WP is molten as specified, and the projectile in a vertical, filling-end-up position.

4.2.4.2 Leakage. After final cleaning, the projectiles shall be placed in an oven in a vertical, filling-end-down position for the 160° ± 5°F method (for the alternate method at 207° to 212°F, they shall be placed in a filling-end-up position) and maintained at the specified temperature for the required period of time, and observed for leakage. When tested at 207° to 212°F, the munition temperature shall be measured by thermacouples and/or thermometers inserted in the burster wells of the test items at random locations. The emission of smoke constitutes leakage.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. Unless otherwise specified in the contract or order the filled and closed shell shall be packed for shipment or storage in accordance with drawing 8837835.

5.2 Marking. In addition to any special marking required by the contract or order, all pallets shall be marked in accordance with drawing 8861520.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification, and any special marking required.

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6.2 Caution. Because of the hazardous nature of WP operations, filling installations are advised to take all necessary precautions required by proper industrial hygiene and safety practices to protect personnel from injury while filling projectiles in accordance with this specification.

6.3 Filling aid. Projectiles filled with approximately 2.0 pounds of WP have been found to comply with 3.2.2. This information is furnished as an aid to procurement and not as a criterion for filling.

6.4 Decontamination. Filled and closed projectiles have been decontaminated by passing through a hot water and detergent spray washer, or by flushing with hot water while scrubbing vigorously with steel wool or a wire brush. Additional cleaning of the burster well has been effected with a steam lance and a power driven wire brush. This information is provided as an aid to production and not as production line design criterion.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1315-A637

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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POSTAGE AND FEES PAID
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OFFICIAL BUSINESS

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